

Tumour ablation for kidney cancer

Tumour ablation uses heat or cold to destroy cancer cells. It may be used to treat small kidney cancers instead of surgery.

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What is tumour ablation?

Tumour ablation means destroying the tumour (cancer). It is sometimes used to treat [kidney cancers](#) smaller than 3 to 4cm.

You may have this treatment if:

- you have a small tumour and are not well enough for [surgery](#).
- you choose not to have surgery
- it is important to try to keep the affected kidney, or you have only 1 kidney
- you have a type of kidney cancer that causes lots of tumours and was passed down from a parent (inherited)
- cancer is affecting both kidneys.

Tumour ablation generally causes fewer side effects and has a quicker recovery time than surgery for kidney cancer. With both surgery and tumour ablation there is a small risk that not all the cancer is removed. This risk is a little higher for tumour ablation. If tumour ablation is an option for you, then your cancer doctor will talk to you about:

- the risks of tumour ablation
- the benefits of avoiding surgery
- the benefits of keeping the kidney.

Types of tumour ablation for kidney cancer

There are different types of tumour ablation. The 2 most common treatments are:

- cryotherapy – this uses liquid nitrogen to freeze the tumour
- radiofrequency ablation (RFA) – this uses an electric current to produce high temperatures to destroy the tumour.

Other types of tumour ablation may be used in [clinical trials](#). These include:

- microwave ablation
- laser ablation
- high-intensity focused ultrasound (HIFU).

Your cancer doctor or specialist nurse will talk to you about which of these methods may be right for you.

You should have a [guided biopsy](#) to collect a tissue sample. This is to give your doctor more information about the cancer. It may be done before tumour ablation, or your doctor may decide to take a sample during treatment.

Having tumour ablation for kidney cancer

Usually, a specialist x-ray doctor will do tumour ablation. They are called a radiologist. You will either be given a sedative to help you relax, or a general anaesthetic.

You may have a catheter put in to drain urine (pee) from your bladder. This is usually removed soon after the procedure.

The doctor uses a local anaesthetic to numb the area around the kidney. Then you have an [ultrasound](#) or [CT scan](#). These scans guide the doctor to the right area of the kidney. The scans also help them monitor what is happening during your treatment.

When the doctor sees the tumour on the monitor, they place 1 or more fine probes through the skin (percutaneously) into the tumour.

Image: The illustration shows a doctor positioning an ablation probe before performing tumour ablation for kidney cancer. It shows a patient lying face down under surgical drapes. An area of skin, above the left kidney, is exposed on the patient's back. A doctor is standing beside the patient. The doctor's right hand is holding the ablation probe above the patient's back, in the kidney area. The doctor's left hand is guiding the tip of the ablation probe into the skin on the patient's back. The doctor is looking at a screen that will show images to help them guide the tip of the probe to the correct place.

The probes freeze or heat the tumour. The extreme temperature destroys the cancer cells. The doctor will also aim to destroy a small area (about 1cm) of healthy tissue around the tumour. This is to try to make sure no cancer cells are left behind to grow back again.

Image: Close-up of cryotherapy Long description: The illustration shows a close-up of cryotherapy for a kidney tumour. It shows a left kidney with a tumour near its centre. 3 ablation probes are shown entering the body through the skin of the left side of the body. The probes enter the kidney and the tips of the probes are in the of the tumour. Freezing gas is shown coming from the tip of the probes.

Keyhole surgery

Sometimes tumour ablation is done using keyhole surgery. You have a general anaesthetic for keyhole surgery.

The surgeon makes a few small cuts in your tummy (abdomen) and passes a [laparoscope](#) through 1 of the cuts to see the tumour. A laparoscope is a thin tube with a light and a camera on the end. It sends video images to a monitor. The surgeon inflates your tummy with gas so that it is easier to see and work with the laparoscope.

Side effects of tumour ablation for kidney cancer

Pain



You will probably have some [pain](#) or discomfort where you had the treatment. Your doctor will give you [painkillers](#) to take regularly for a few days.

Bloating



If you had gas in your tummy for the laparoscope, you may feel bloated and have some discomfort in your shoulders. This improves over a few days as your body absorbs the gas.

Feeling sick



Sometimes people [feel sick](#) immediately after tumour ablation. If this happens, tell your nurse or doctor. They can give you anti-sickness drugs.

Raised temperature



You may feel a little unwell for the first few days and have a slightly raised temperature. You will probably also [feel tired](#). Drinking plenty of fluids will help. If your temperature does not return to normal, or if it goes above 38°C (100.4°F), contact your doctor. This may be caused by an [infection](#).

Blood in your urine



You may notice some blood in your urine (pee). This should go away after a few days.

You usually need to stay in a hospital bed for 4 to 6 hours after the treatment.

Possible complications after tumour ablation

The risk of complications after tumour ablation is low.

Possible complications include:

- infection – you may be given antibiotics to reduce the risk of this happening
- bleeding – you will be monitored during treatment and for a few hours afterwards
- a narrowing of the [ureter](#) – this can affect how urine drains from the kidney.

You will have a scan after treatment to check for any complications. These can be treated straight away if needed.

Going home after tumour ablation

After tumour ablation, you may go home on the same day or the day after treatment. This depends on how quickly you recover.

Your surgeon or nurse can tell you when you can start doing everyday activities again.

Before you leave hospital, you will be given an appointment for a check-up. This will be at an outpatient clinic. Your doctor or nurse may arrange for you to have a [CT scan](#) to see the result of the treatment.

You will have regular follow-up scans to check the kidney for any signs of the cancer growing back.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our kidney cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Powles T, Albiges L, Bex A, et al. Renal cell carcinoma: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Ann Oncol. 2024 May 22. Available from: <https://doi.org/10.1016/j.annonc.2024.05.537>

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help